

Unit-II

2.5 SQL Joins

SQL joins are fundamental tools for combining data from multiple tables in relational databases.

- For example, consider two tables where one table (say Student) has student information with id as a key and other table (say Marks) has information about marks of every student id. Now to display the marks of every student with name, we need to join the two tables.
- Please remember, we store data into multiple tables as part of database normalization to avoid anomalies and redundancies.

Types of SQL Joins

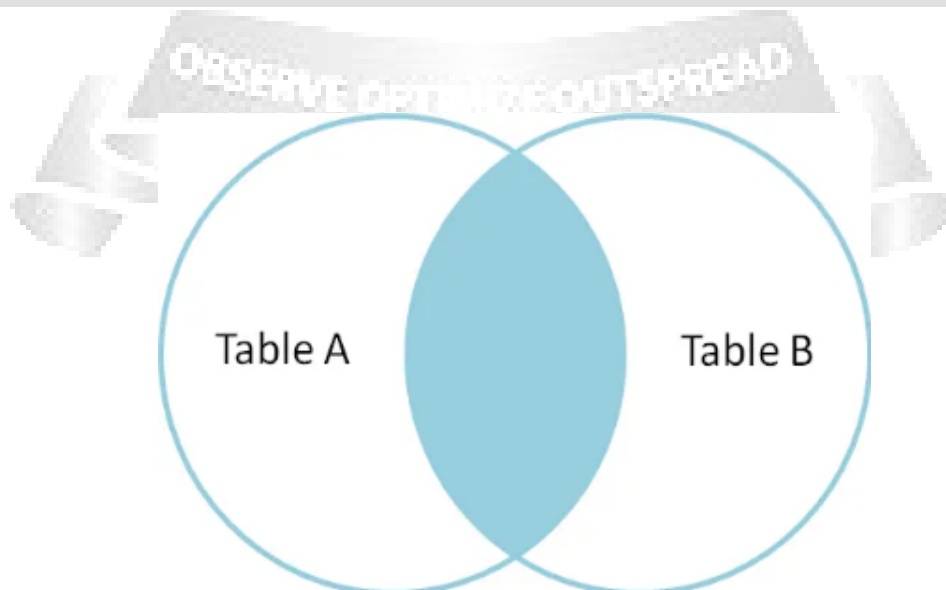
Let us visualize how each join type operates:

1. SQL INNER JOIN

The INNER JOIN keyword selects all rows from both the tables as long as the condition is satisfied. This keyword will create the result set by combining all rows from both the tables where the condition satisfies i.e value of the common field will be the same.

Syntax:

```
SELECT table1.column1,table1.column2,table2.column1,....  
  
FROM table1 INNER JOIN  
table2 ON table1.matching_column = table2.matching_column;
```



Example of INNER JOIN

Consider the two tables, Student and StudentCourse, which share a common column ROLL_NO. Using SQL JOINS, we can combine data from these tables based on their relationship, allowing us to retrieve meaningful information like student details along with their enrolled courses.

1. Student Table:

ROLL_NO	NAME	ADDRESS	PHONE	AGE
1	HARSH	DELHI	XXXXXXXXXX	18
2	PRATIK	BIHAR	XXXXXXXXXX	19
3	RIYANKA	SILIGURI	XXXXXXXXXX	20
4	DEEP	RAMNAGAR	XXXXXXXXXX	18
5	ANITA	MUMBAI	XXXXXXXXXX	21

2. StudentCourse Table:

COURSE_ID	ROLL_NO
1	1
2	2
3	3
6	7

Let's look at the example of INNER JOIN clause, and understand it's working. This query will show the names and age of students enrolled in different courses.

Query:

```
SELECT StudentCourse.COURSE_ID, Student.NAME, Student.AGE FROM Student
INNER JOIN StudentCourse
ON Student.ROLL_NO = StudentCourse.ROLL_NO;
```

Output:

COURSE_ID	NAME	AGE
1	HARSH	18
2	PRATIK	19

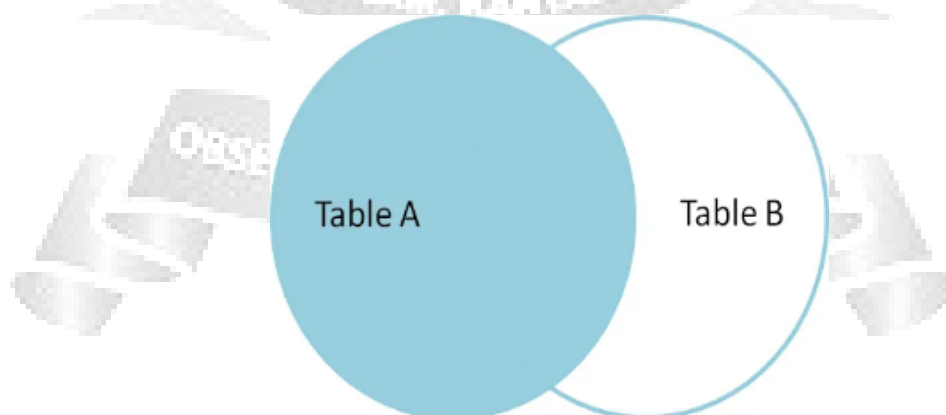
COURSE_ID	NAME	AGE
3	RIYANKA	20

2. SQL LEFT JOIN

A LEFT JOIN returns all rows from the left table, along with matching rows from the right table. If there is no match, NULL values are returned for columns from the right table. LEFT JOIN is also known as LEFT OUTER JOIN.

Syntax

```
SELECT table1.column1,table1.column2,table2.column1,...
FROM table1
LEFT JOIN table2
ON table1.matching_column = table2.matching_column;
```

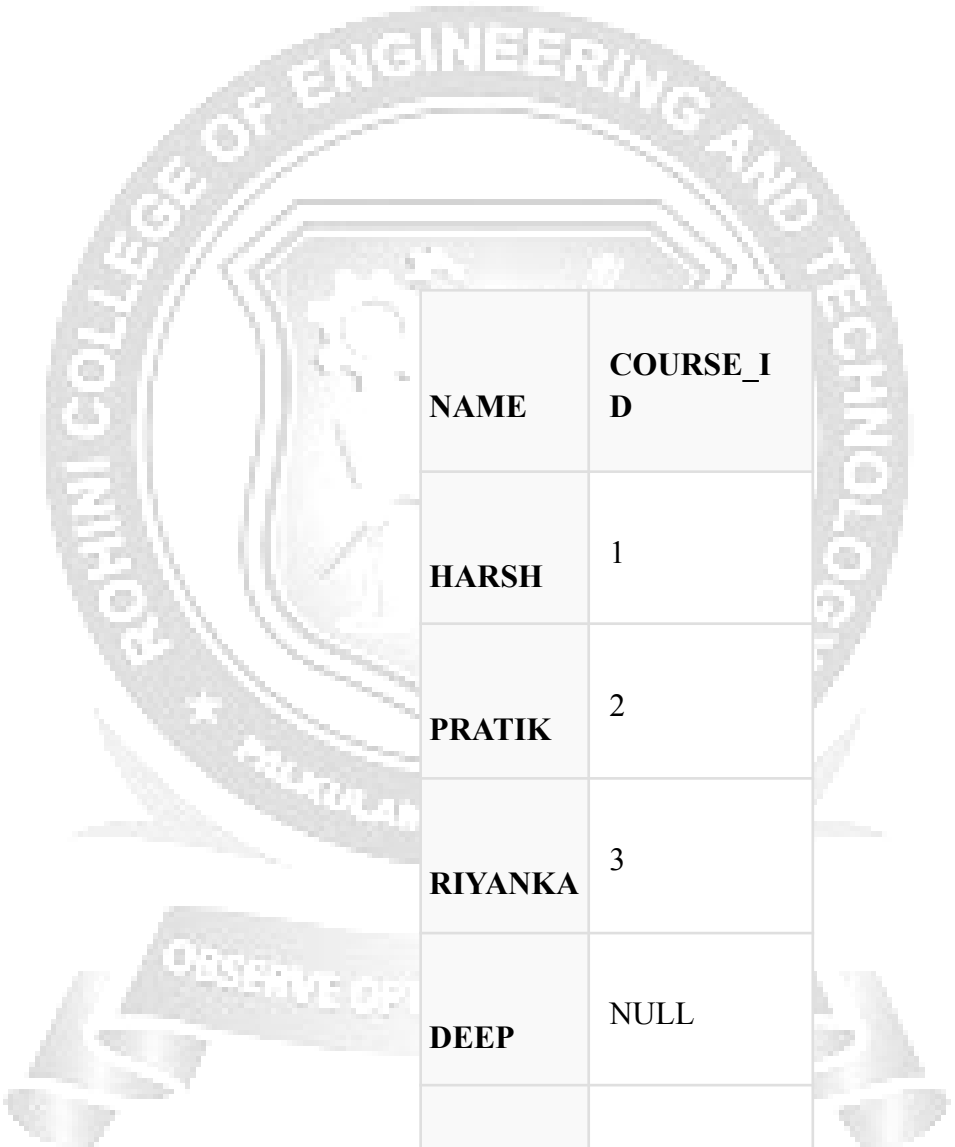


LEFT JOIN Example

In this example, the LEFT JOIN retrieves all rows from the Student table and the matching rows from the StudentCourse table based on the ROLL_NO column.

Query:

```
SELECT Student.NAME, StudentCourse.COURSE_ID
FROM Student
LEFT JOIN StudentCourse
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output:

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	3
DEEP	NULL
ANITA	NULL

3. SQL RIGHT JOIN

RIGHT JOIN returns all the rows of the table on the right side of the join and matching rows for the table on the left side of the join. It is very similar to LEFT JOIN for the rows for which there is no matching row on the left side, the result-set will contain null. RIGHT JOIN is also known as RIGHT OUTER JOIN.

Syntax

```
SELECT table1.column1,table1.column2,table2.column1,....  
FROM table1  
RIGHT JOIN table2  
ON table1.matching_column = table2.matching_column;
```

Key Terms

- **table1**: First table.
- **table2**: Second table
- **matching_column**: Column common to both the tables.



RIGHT JOIN Example

In this example, the RIGHT JOIN retrieves all rows from the StudentCourse table and the matching rows from the Student table based on the ROLL_NO column.

Query:

```
SELECT Student.NAME,StudentCourse.COURSE_ID  
FROM Student  
RIGHT JOIN StudentCourse  
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output:

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	3
NULL	6

4. SQL FULL JOIN

FULL JOIN creates the result-set by combining results of both LEFT JOIN and RIGHT JOIN. The result-set will contain all the rows from both tables. For the rows for which there is no matching, the result-set will contain NULL values.

